

Ex.No :7 Use AF Logical OSE to extract data from an Android device Date

Date:-

Aim:-

To use AFLogical OSE to perform a logical extraction of data (such as contacts, SMS, call logs, and device properties) from an Android device via the ADB interface, without requiring root access, for the purpose of mobile/digital forensic investigation.

Description:-

AFLogical OSE (Open Source Edition) is a tool used to extract data from Android devices. It's part of the Open Source Android Forensics project and is often used for logical extraction, which means pulling data like contacts, messages, and call logs without directly accessing the file system. Here's a step-by-step guide to using AFLogical OSE to extract data from an Android device.

Step 1: Prepare Your Environment

1. Download AFLogical OSE:

- Visit the [AFLogical OSE GitHub page](#) and download the latest release.
Alternatively, you can clone the repository using Git.

2. Install Java (if not already installed):

- AFLogical OSE requires Java to run. Make sure you have Java installed on your machine. You can download it from the [official Java website](#).

3. Install Android Debug Bridge (ADB):

- ADB is a command-line tool that allows communication with Android devices.
Download it from the [Android Developer's website](#) and install it on your system. ○
Add ADB to your system's PATH environment variable to easily run ADB commands from the terminal or command prompt.

4. Enable USB Debugging on the Android Device:

- On the Android device, go to Settings > About Phone and tap Build Number seven times to enable Developer Options.
- Go to Settings > Developer Options and enable USB Debugging.

Step 2: Connect the Android Device to Your Computer

1. Connect via USB: ○ Use a USB cable to connect the Android device to your computer.

2. Verify ADB Connection:

- Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:

```
Bash    adb            devices
```

- This command lists connected devices. If your device is listed, you are ready to proceed. If not, ensure that USB Debugging is enabled and that the correct drivers are installed.

Step 3: Extract Data Using AFLogical OSE

1. Run AFLogical OSE:

- Navigate to the directory where you have downloaded or extracted AFLogical OSE.

2. Push the APK to the Android Device:

- Run the following command to install the AFLogical OSE APK on the device:

bash

```
adb install aflogical.apk
```

- This installs the AFLogical app on the device, which will be used to collect data.

3. Launch the AFLogical OSE App on the Device:

- On the Android device, open the AFLogical app.
- You'll see options to extract various types of data (e.g., contacts, SMS, MMS, call logs). Select the data types you want to extract.

4. Start Data Extraction:

- Once you've selected the data types, the app will start the extraction process. ○ Data is stored in .csv files on the device's storage, typically in a directory named aflogical.

Step 4: Transfer Extracted Data to Your Computer

1. Use ADB to Pull Data:

- After extraction, you need to transfer the data files from the Android device to your computer. Run:

bash

```
adb pull /sdcard/aflogical /path/to/destination
```

- Replace /path/to/destination with the directory path where you want to save the extracted data on your computer.

2. Verify the Data:

- Navigate to the destination directory and check the .csv files to ensure all required data has been extracted.

Step 5: Analyze the Extracted Data

1. Open the CSV Files:

- Use Excel, Google Sheets, or any text editor to open and analyze the .csv files.

These files contain the extracted data like contacts, SMS, MMS, and call logs.

2. Review and Document:

- Carefully review the data for any relevant information. Document your findings and consider exporting the data into a report format for further analysis.

Step 6: Clean Up

1. Uninstall AFLogical OSE:

- Once data extraction is complete, you can uninstall the AFLogical OSE app from the Android device by running:

```
bash
```

```
adb uninstall com.viaforensics.android.aflogical
```

2. Disconnect the Device:

- Safely disconnect the Android device from your computer.

By following these steps, you can use AFLogical OSE to effectively extract and analyze data from an Android device. This process is valuable for forensic investigations, especially when dealing with logical data that doesn't require root access to the device.

OUTPUT :

Installing JAVA :

```
C:\Users\UDAY KUMAR REDDY>java -version
openjdk version "25.0.4.1" 2026-08-18 LTS
OpenJDK Runtime Environment Temurin-25.0.4.1+1 (build 25.0.4.1+1-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.4.1+1 (build 25.0.4.1+1-LTS, mixed mode, sharing)

C:\Users\UDAY KUMAR REDDY>
```

Installing ADB Tools :

```
C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>adb version
Android Debug Bridge Version 1.0.41
Version 37.0.1-15733141
Installed as C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools\adb.exe
Running on Windows 10.0.26200

C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>
```

ADB Device Connection :

```
C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>adb devices
* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached
ZD2227X3N4      device

C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>
```

Installing alfllogical.apk :

```
C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>adb install
com.viaforensics.android.aflogical_ose_1.5.2_OSE_152.apk
Performing Streamed Install
Success
C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>
```

Capturing SMS and Call Logs :

```
C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>adb.exe shell content query --uri
content://sms > sms.txt

C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>adb.exe shell content query --uri
content://call_log/calls > call_logs.txt

C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>
```

Extracted SMS Logs :

```
Row: 0 id=6810, thread_id=2294, address=919032675630, person=NULL, date=1788759138655, date_sent=1788759138000, protocol=0, read=0, status=-1, type=1, reply_path_present=1,
subject=proto:CjokImWv55nb29n6dUuYwSkcn9pZC5hCHBzLn1lc3NhZ2Uy4sFCIAK0hR9nBzciDFFtPPTKS5JVNROAIMIM, body=Dear Customer, You have a missed call from +919032675630
The last missed call was at 10:58 AM on 07-Sep-2026 Thankyou, Team Jio., service_center=+919014308414, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.
messaging, seen=1

Row: 2 id=6808, thread_id=1807, address=JX-ISESVS-S, person=NULL, date=1788684739072, date_sent=1788644785000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,
subject=proto:CjokImWv55nb29n6dUuYwSkcn9pZC5hCHBzLn1lc3NhZ2Uy4sFCIAK0hR9nBzciDFFtPPTKS5JVNROAIMIM, body=GROMINVESTTECHPRIVATE LIMITED on 29-08-2026 reported
your Fund bal Rs.0.000 & Securities bal 0.000. This excludes your Bank, DP & PMS bal with the broker-NSE, service_center=+9170030785026, locked=0, sub_id=2, error_code=-1,
creator=com.google.android.apps.messaging, seen=1

Row: 2 id=6808, thread_id=2237, address=JX-ISMSA-S, person=NULL, date=1788533923394, date_sent=1788533919000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,
subject=proto:CjokImWv55nb29n6dUuYwSkcn9pZC5hCHBzLn1lc3NhZ2Uy4sFCIAK0hR9nBzciDFFtPPTKS5JVNROAIMIM, body=PNR-4444128174
Trn:22715
Dt:05-09-26 Dep.Time-09:47 Hrs.
From KRNT to NDU
Cls:SL
P1-S4,39
P2-S4,29
P3-S4,32
Boarding allowed from KRNT only
Chart Prepared
URL for coach position: https://enquiry.indianrail.gov.in/mntes/c?u=ggIfjN9d7NrejngkW8I
Download RailOne for latest updates https://railmone.indianrailways.gov.in
For Enquiry/Complaint/Assistance, please dial 139 IR-CRIS, service_center=+917019075011, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 3 id=6807, thread_id=1771, address=JM-NISESVS-S, person=NULL, date=17885306637133, date_sent=1788530635000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,
subject=proto:CjokImWv55nb29n6dUuYwSkcn9pZC5hCHBzLn1lc3NhZ2Uy4sFCIAK0hR9nBzciDFFtPPTKS5JVNROAIMIM, body=Beware while dealing based on unsolicited tips through
WhatsApp,Telegram,SMS,calls,YouTube,etc,& take an informed decision. Visit https://bit.ly/3qQx5Co -NSE, service_center=+917021079516, locked=0, sub_id=2, error_code=-1,
creator=com.google.android.apps.messaging, seen=1

Row: 4 id=6806, thread_id=1806, address=JA-JIOPAY-S, person=NULL, date=1788522742702, date_sent=1788522691000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,
subject=proto:CjokImWv55nb29n6dUuYwSkcn9pZC5hCHBzLn1lc3NhZ2Uy4sFCIAK0hR9nBzciDFFtPPTKS5JVNROAIMIM, body=Jio
On 04-Sep-26 17:21, a data pack of 1GB has been successfully activated.
Jio Number: 8074999510
Data Pack Details: 2 GB
Validity: till 05-Sep-26 00:27 (midnight).
You have consumed 64 Kbps of data.
For best internet experience, use 4G or higher network (64 Kbps speed is slow).
To check your data pack details, visit: http://tiny.jio.com/dbuydatapack
```

Extracted Call Logs :

```

Row: 0 formatted_number=NULL, duration=59, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=3, photo_id=0, post_dial_digits=,
call_screening_app_name=NULL, priority=0, number=, countryiso=IN, cnap_name=, photo_uri=NULL, geocoded_location=India, missed_reason=0, block_reason=0,
cached_convert_number=NULL, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1,
numbertype=NULL, features=68, transcription=NULL, last_modified=1729700812435, _id=8385, new=0, date=1718092698184, name=Seenu, type=1, presentation=1,
via_number=, numberlabel=NULL, normalized_number=, composer_photo_uri=NULL, phone_account_address=, phone_account_hidden=0, lookup_uri=NULL, voicemail_uri=NULL,
matched_number=NULL, transcription_state=0, data_usage=NULL, location=NULL, call_screening_component_name=NULL, is_read=1
Row: 1 formatted_number=NULL, duration=0, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=NULL, photo_id=0, post_dial_digits=,
call_screening_app_name=NULL, priority=0, number=, countryiso=IN, cnap_name=NULL, photo_uri=NULL, geocoded_location=India, missed_reason=0, block_reason=0,
cached_convert_number=NULL, subscription_component_name=NULL, add_for_all_users=1, numbertype=NULL, features=0, transcription=NULL, last_modified=1729700812435,
_id=8417, new=0, date=1718254552633, name=, type=2, presentation=1, via_number=, numberlabel=NULL, normalized_number=, composer_photo_uri=NULL,
phone_account_address=NULL, phone_account_hidden=0, lookup_uri=NULL, voicemail_uri=NULL, matched_number=NULL, transcription_state=0, data_usage=NULL,
location=NULL, call_screening_component_name=NULL, is_read=1
Row: 2 formatted_number=+91 90520 33059, duration=0, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=3, photo_id=0, post_dial_digits=,
call_screening_app_name=NULL, priority=0, number=, countryiso=IN, cnap_name=, photo_uri=, geocoded_location=India, missed_reason=1114112, block_reason=0,
cached_convert_number=NULL, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1,
numbertype=64, transcription=NULL, last_modified=1729700812435, _id=8461, new=0, date=1718982030862, name=Sami, type=3, presentation=1, via_number=,
numberlabel=Mobile, normalized_number=, composer_photo_uri=NULL, phone_account_address=, phone_account_hidden=0,
lookup_uri=content://com.android.contacts/contacts/lookup/188i373dd0188e90969c/4201?directory=0, voicemail_uri=NULL, matched_number=NULL, transcription_state=0,
data_usage=NULL, location=NULL, call_screening_component_name=NULL, is_read=1
Row: 3 formatted_number=+91 77991 68016, duration=38, subject=NULL, is_call_log_phone_account_migration_pending=0, subscription_id=3, photo_id=0,
post_dial_digits=, call_screening_app_name=NULL, priority=0, number=+917799168016, countryiso=IN, cnap_name=+917799168016, photo_uri=, geocoded_location=India,
missed_reason=0, block_reason=0, cached_convert_number=NULL,
subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, add_for_all_users=1, numbertype=0, features=64,
transcription=NULL, last_modified=1729700812435, _id=8499, new=0, date=1720318909704, name=, type=1, presentation=1, via_number=, numberlabel=Mobile,
normalized_number=+917799168016,

```

Extracted Device Properties :

```

PS C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>
Write-Host "Device Model:" (.adb.exe shell getprop ro.product.model)
Device Model: moto g34

PS C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>
Write-Host "Android Version:" (.adb.exe shell getprop ro.build.version.release)
Android Version: 14

PS C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>
Write-Host "Security Patch Level:" (.adb.exe shell getprop ro.build.version.security_patch)
Security Patch Level: 2025-06-05

PS C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>

```

Uninstalling aflogical.apk :

```

C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>.adb.exe
uninstall --user 0
com.viaforensics.android.aflogical_ose_1.5.2_OSE_152.apk

C:\Users\UDAY KUMAR REDDY\Downloads\platform-tools-latest-windows\platform-tools>

```

Rubrics:-

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result :-

The experiment successfully performed a logical extraction of data from the Android device using the A-FLogical OSE framework via ADB interface connection. The tool bypassed file system access limitations to cleanly pull device properties, **SMS logs, and call records into structured .csv files**, which were verified on the local host before safely uninstalling the APK tool.